Borines, Bucago, Datiles, Desio, Maido
BSIS 4A
Task 1

ML Project Proposal

Predicting Livestock Health Problems: Early Identification of Likely Diseases

Machine Learning Project | Task 1
Members: Borines, Rejelyn; Bucago, Princess Kyle; Datiles, Ma. Christine; Desio, Juami Shaniqua; Maido, Rich Dave

1. Dataset

- **Dataset:** Livestock Symptoms and Diseases, a public dataset on Kaggle by researcher1548 [1]. It has been used in published research [2].
- **Licence:** Publicly available on Kaggle and cited as required by its terms.
- **Size:** 43,778 records. Each row is one animal case (one animal's recorded signs and diagnosis); each column is one recorded detail.
- **Features:** animal species (cow, buffalo, sheep or goat), age, body temperature, and three observed symptoms.
- **Label:** the confirmed disease for each animal: anthrax, blackleg, foot and mouth disease, lumpy skin disease, or pneumonia.
- **Privacy:** The data describes animals only and has no personal information. Results will be checked per species so the model is not accurate for one species only.

2. Problem Statement

Farmers may not notice livestock illness early, and delays increase animal losses and disease spread. Can age, species, body temperature and observed symptoms predict which disease an animal most likely has, so a farmer can isolate it and call a veterinarian sooner?

- **Type:** multi-class classification (five diseases).
- **Target:** the confirmed disease recorded for each animal.

3. Metrics

Our main score is macro F1, which treats all five diseases equally. Supporting scores are recall for each disease, balanced accuracy, and one-vs-rest ROC AUC, with a confusion matrix. Missing a serious or contagious disease such as anthrax is worse than a false alarm, so recall for each disease matters most. We will report the mean and spread across folds.

4. Baseline and Plan

- **Baseline:** always predict the most common disease. This is expected to give about one correct answer in five and a macro F1 near zero. The exact value will be computed in the notebook.
- **Models:** logistic regression, random forest, and k-nearest neighbours or SVM.
- **Testing:** stratified 80/20 split, then 5-fold cross-validation on the training part. The test set is used only once, at the end.

References

[1] researcher1548, "Livestock Symptoms and Diseases," Kaggle. [Online]. Available: https://www.kaggle.com/datasets/researcher1548/livestock-symptoms-and-diseases

[2] N. Naik, Jagadisha N., and S. Ramu, "Buffalo disease diagnosis using machine learning: A symptom-based text classification approach," *J. Buffalo Sci.*, vol. 14, pp. 65-73, 2025, doi: 10.6000/1927-520X.2025.14.07.